The University of Faisalabad

# Network Model

## Types of Network:
## Personal Area Network,
## Local Area Network
## Metropolitan Area Network,
## Wide Area Network
## Piconet, Ethernet, Fast Ethernet, Metro Ethernet, Internetwork

Lecture # 7 and 8

Week# 4

Data Communications and Networking, Fourth Edition, Behrouz A. Forouzan

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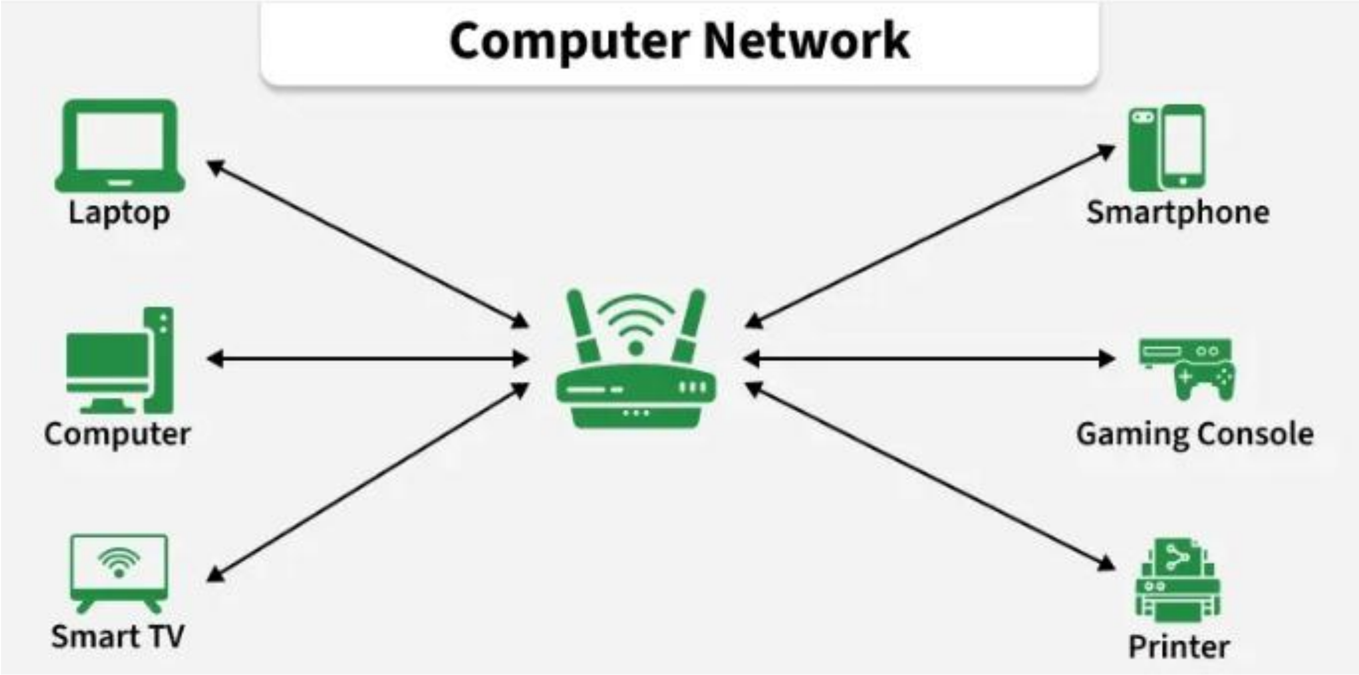
# Computer Networks

- A computer network is a system of interconnected devices, such as computers, servers, smartphones, and printers, that communicate to exchange data. They communicate using wired or wireless connections and can range from small home networks to the global Internet
- **Simple Example:**
- Computers in a school lab connected to one printer
- Mobile phones connected to the same Wi-Fi
- ATMs connected to bank servers
- All these are **computer networks**

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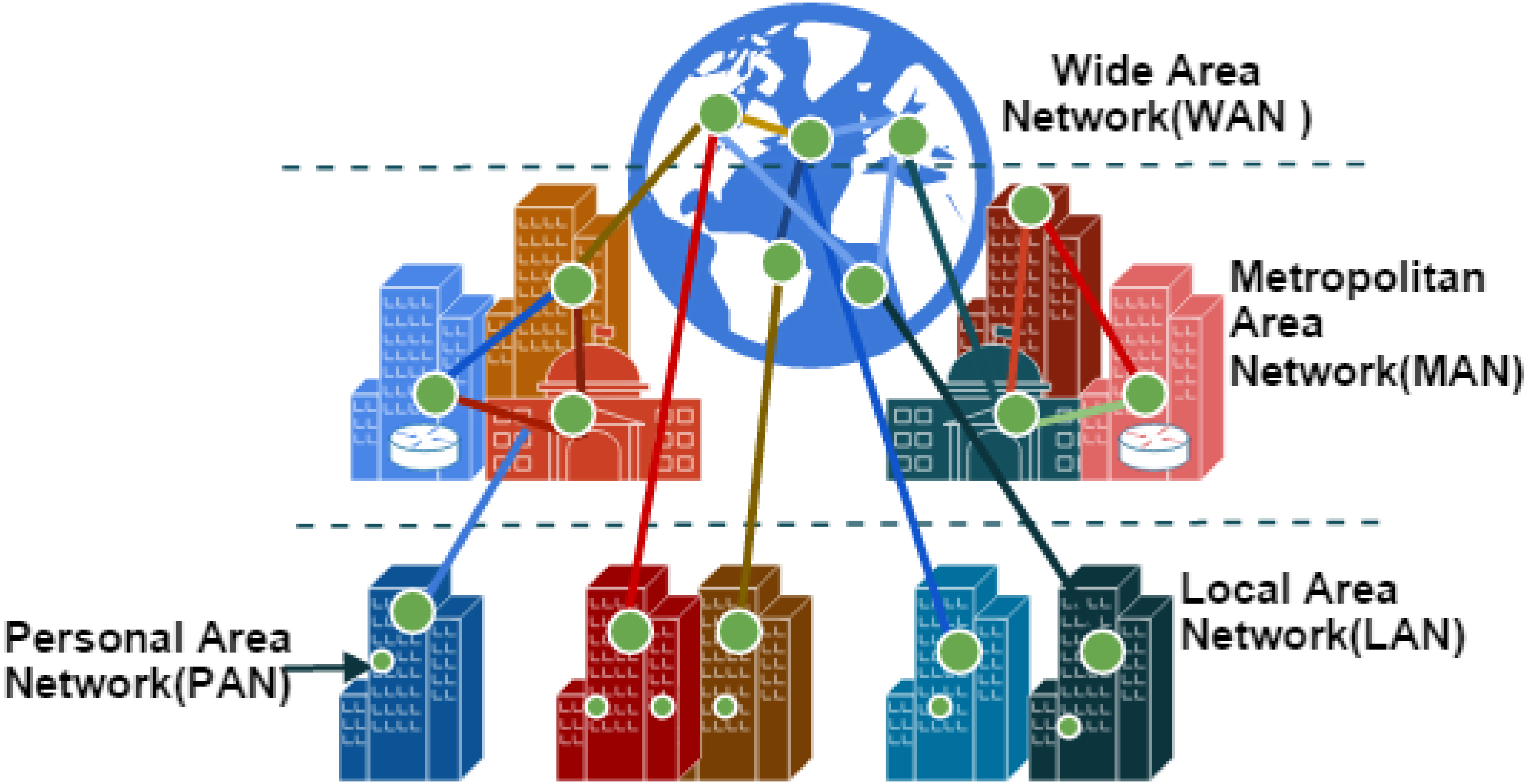
## Types of Computer Networks

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# Personal Area Network (PAN)

- **What is PAN?**
- A **Personal Area Network** is a **small network used by one person** to connect personal devices.

**Range:** Few meters (around a room)

**How PAN Works (Easy)**

- Devices connect using **Bluetooth.**
- One device usually controls others

**Examples**

- Mobile phone connected to **Bluetooth earbuds**
- Laptop connected to **wireless mouse**
- Phone sharing internet via **hotspot**

**Real-Life Example**

- Your phone + smartwatch + earbuds = **PAN**

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# Personal Area Network (PAN)




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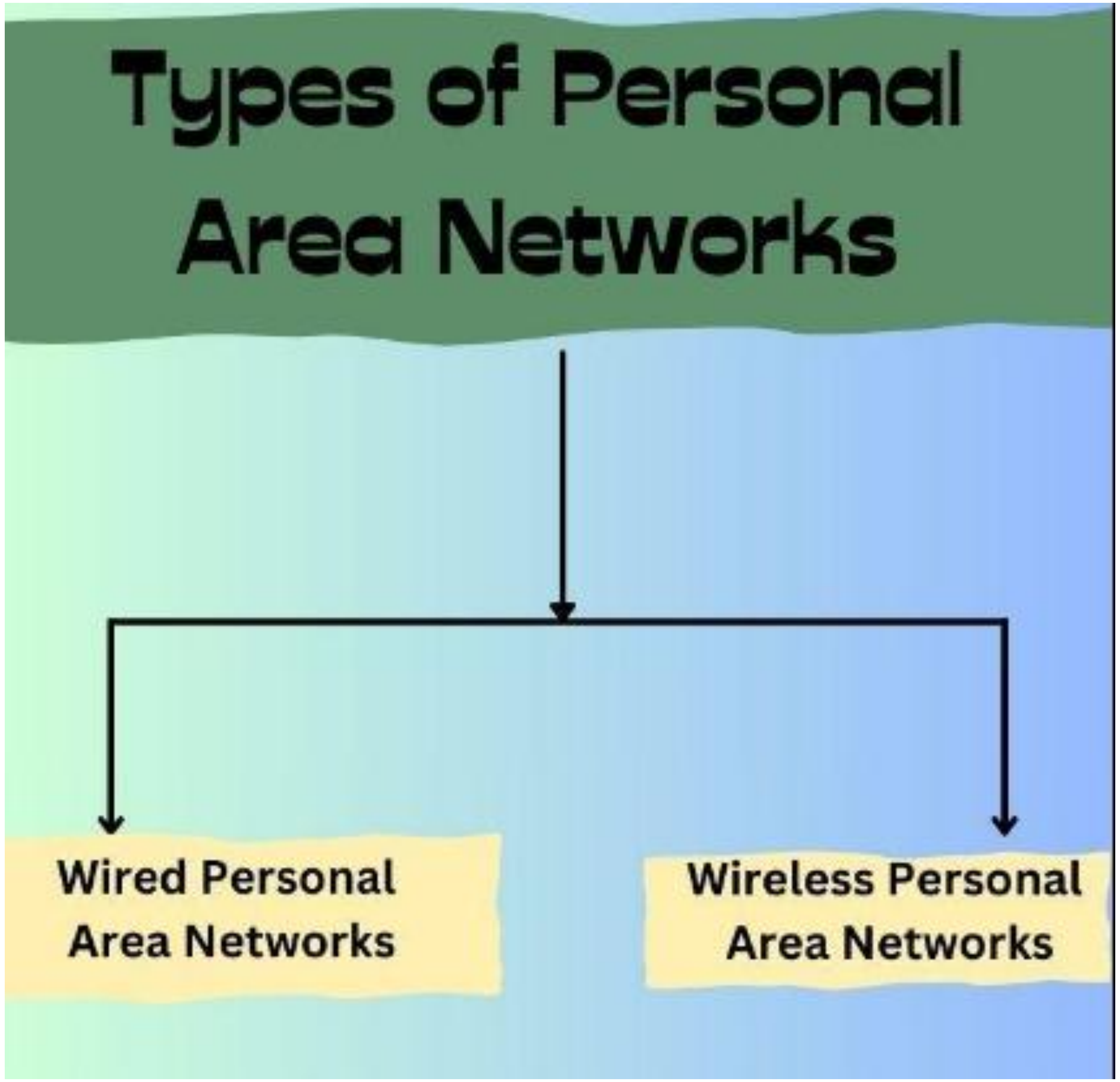
# Types of Personal Area Network



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# Wired PAN

A **Wired PAN** connects personal devices using **physical cables**.

## How Wired PAN Works

- Devices are connected using **data cables**
- Data travels through **wires**
- Very stable connection

## Examples

- Mobile phone connected to laptop via **USB cable**
- Printer connected to computer using **USB**
- External hard drive connected to PC

## Real-Life Example

- Connecting your phone to a laptop using a charging/data cable to transfer photos.

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# Wireless PAN (WPAN)

**Wireless PAN** connects devices **without cables**, using **wireless technologies**.

**Technology Used**

- Bluetooth

**How Wireless PAN Works**

- Devices communicate using **radio waves**
- One device may act as **controller**
- Short-range communication

**Examples**

- Bluetooth headphones connected to phone
- Wireless mouse and keyboard
- Smartwatch connected to mobile

**Real-Life Example**

- Phone connected to Bluetooth earbuds = **Wireless PAN**

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# Local Area Network (LAN)

**Local Area Network** connects computers **within a small area**, like:

- Home
- Office
- School
- Lab
- **Range:** Building or campus

## How LAN Works

- Devices connect through **Ethernet cables or Wi-Fi**
- A **switch or router** manages data flow

## Examples

- Computer lab in a university
- Office computers sharing printer
- Home Wi-Fi network

## Real-Life Example

- All computers in your department lab connected together = **LAN**

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# Local Area Network (LAN)



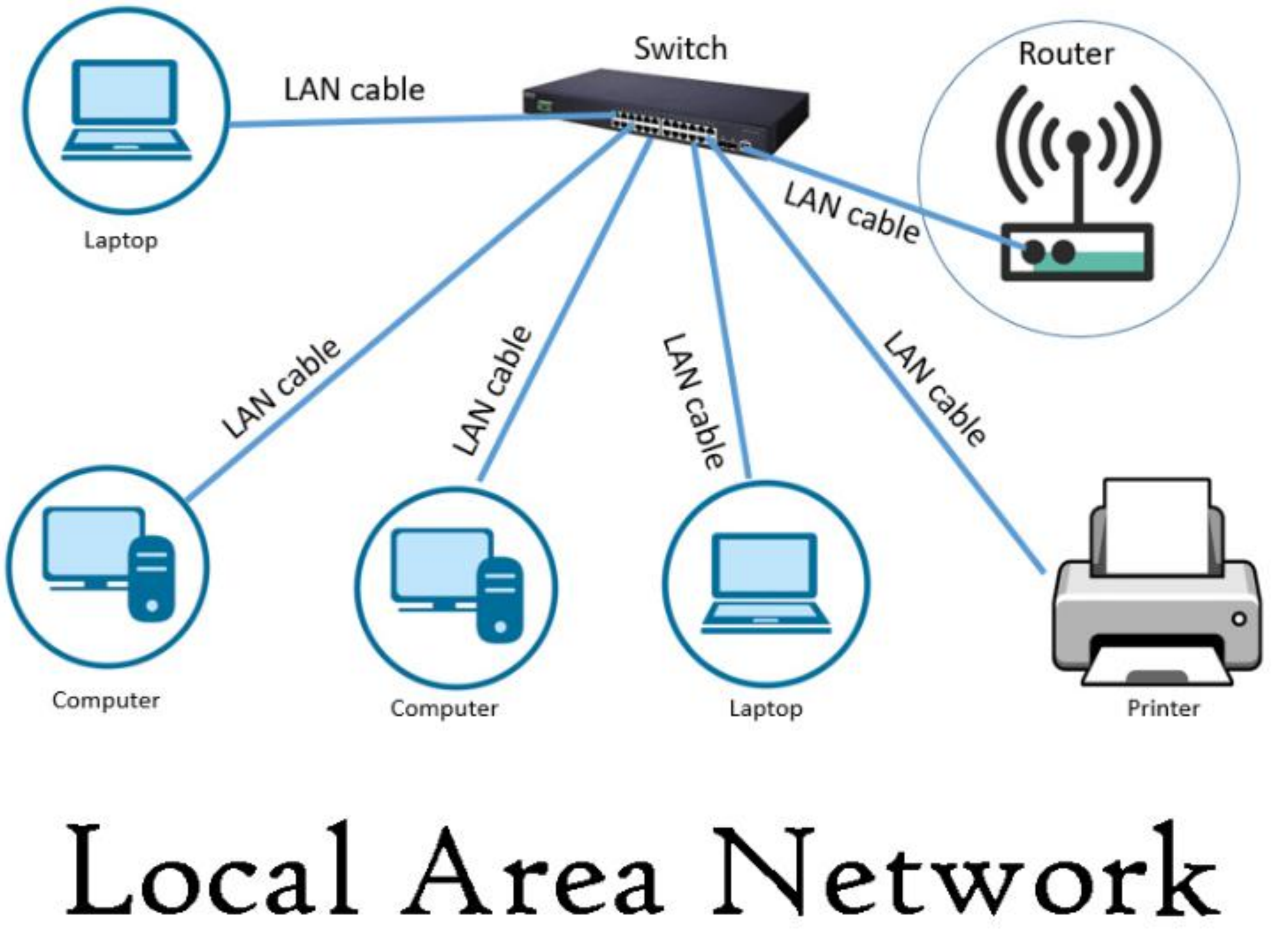


Diagram of local area network

# Metropolitan Area Network (MAN)



**Metropolitan Area Network** connects **multiple LANs within a city**.

**Range:** City or large campus area

**How MAN Works**

- Uses **fiber optic cables**
- Usually managed by **large organizations**

**Examples**

- University campuses across a city
- Cable TV networks
- City-wide Wi-Fi

**Real-Life Example**

- All branches of a university in one city connected = **MAN**

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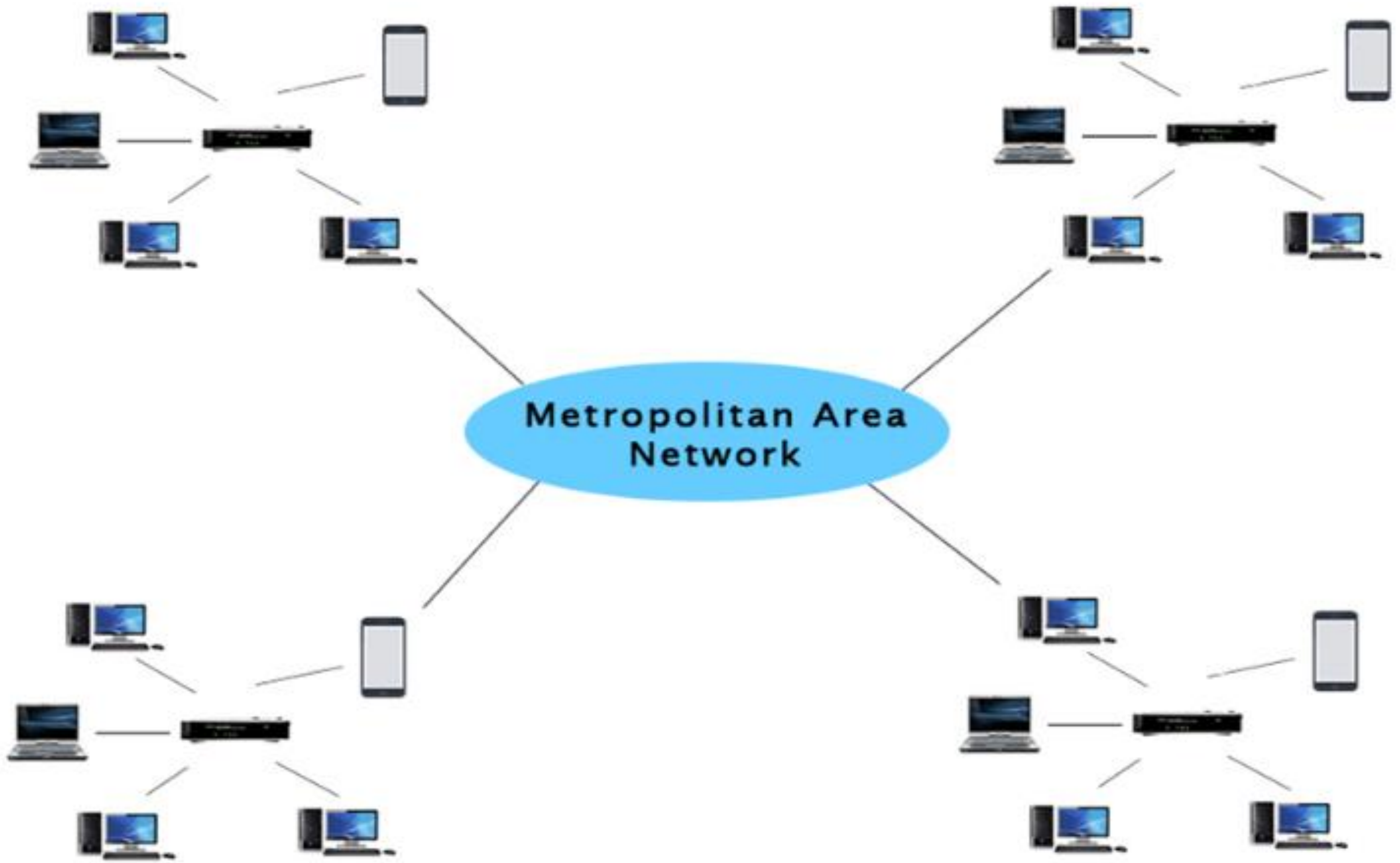
# Metropolitan Area Network (MAN)

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# Wide Area Network (WAN)



**Wide Area Network** connects networks **across countries or the world**.

**Range:** Country to worldwide

**How WAN Works**

- Uses **satellites, undersea cables, fiber**
- Routers forward data over long distances

**Examples**

- WhatsApp Message worldwide
- YouTube Video
- Online Shopping

**Real-Life Example**

- You accessing Google servers from Pakistan = **WAN**

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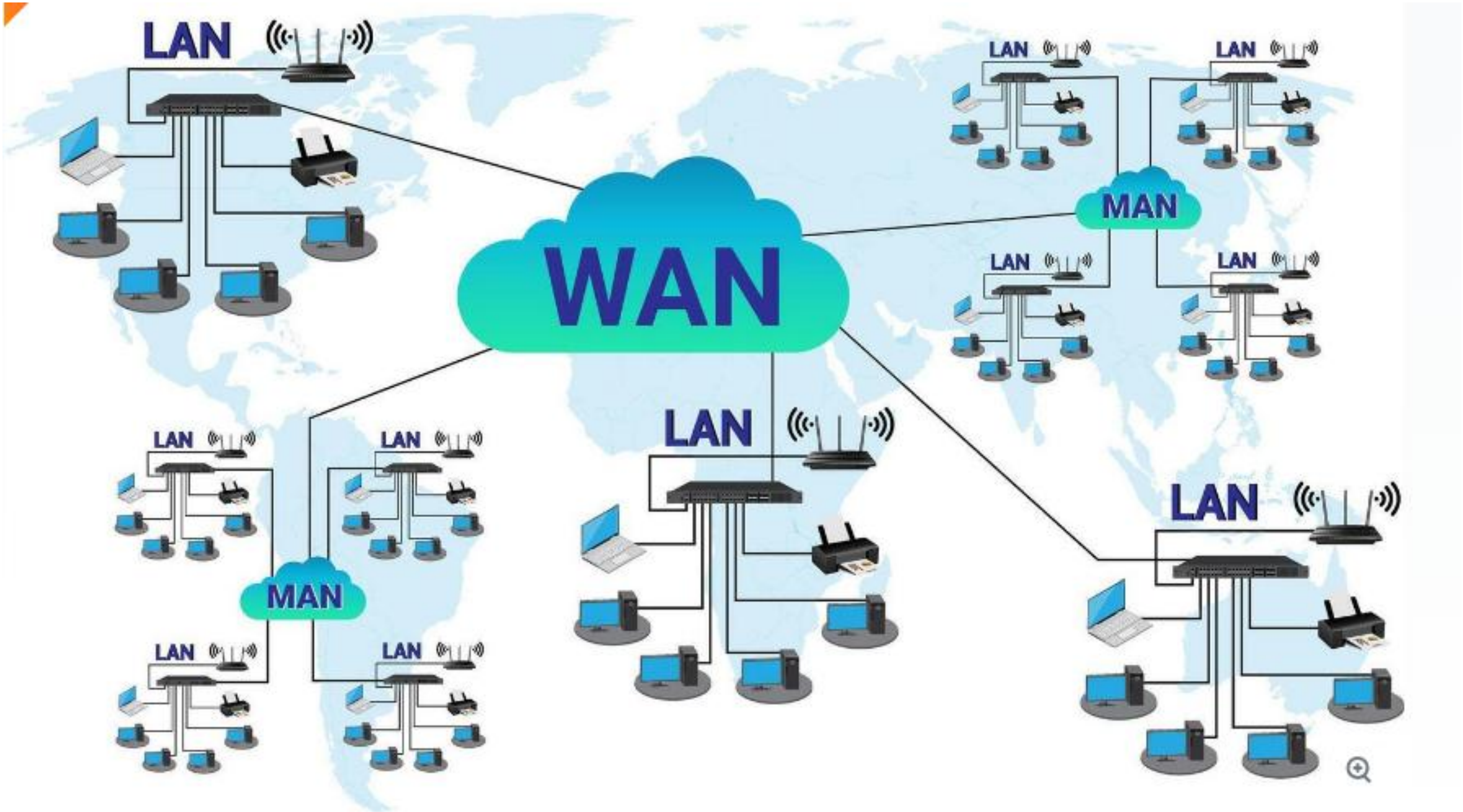
# Wide Area Network (WAN)



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# What is a Piconet?

- A **Piconet** is a **very small wireless network** made using **Bluetooth**.
- **Piconet = One main Bluetooth device + a few nearby Bluetooth devices**

**Master Device**

- The **main device**
- Controls communication
- Decides **who can talk and when**

**Slave Devices**

- Other devices connected to the master
- Follow instructions from master

**Rule:**

- **1 master**
- **Up to 7 slave devices**

**Example**

You connect:

Bluetooth earbuds

Smartwatch

Car Bluetooth

All connected to **one phone**

This group is called a **Piconet**.

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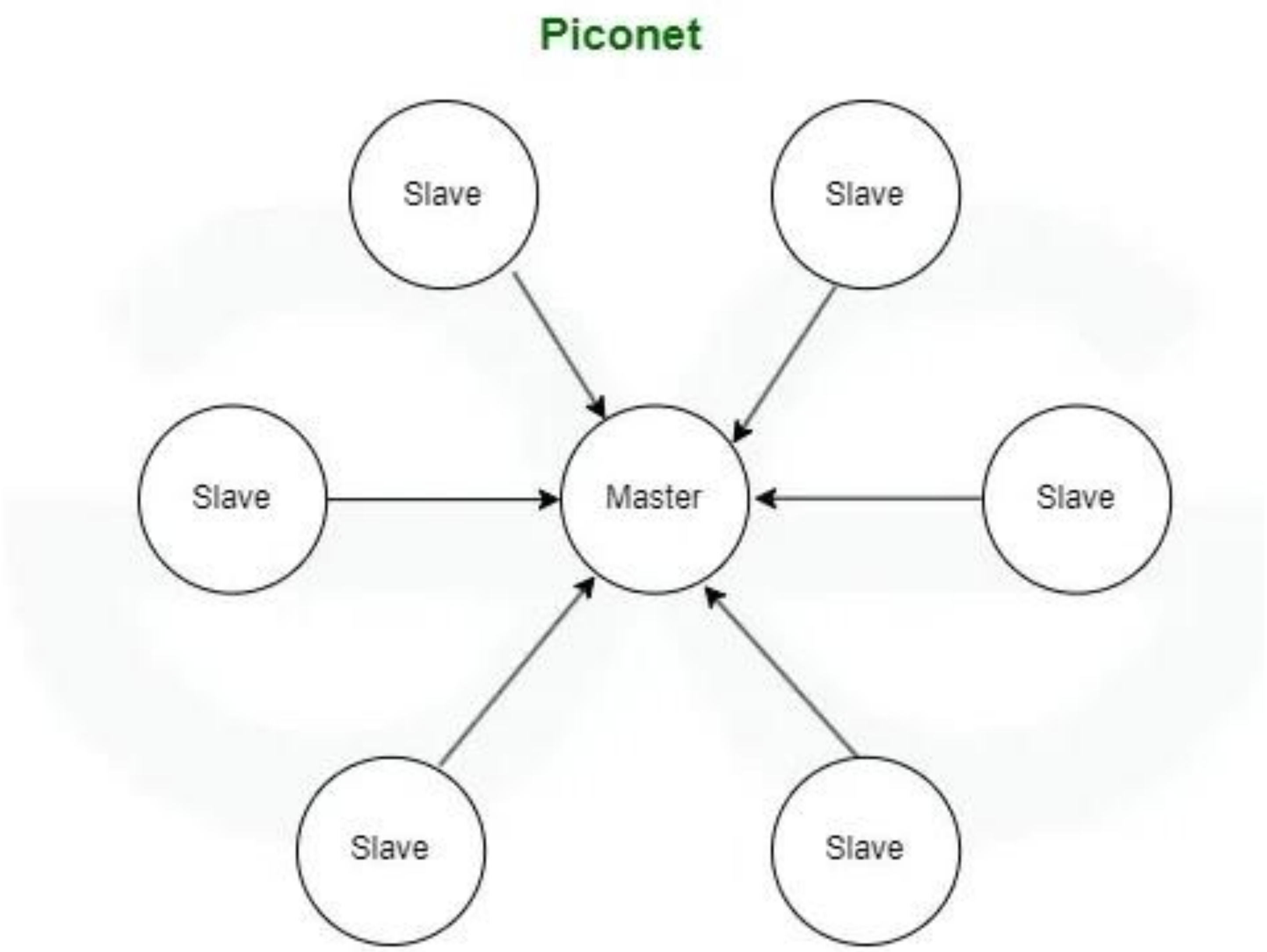
# Piconet



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# Ethernet

**Ethernet** is the most common **wired LAN technology**.

## How Ethernet Works

- Uses **Ethernet cables**
- Data sent in frames
- Uses MAC addresses

## Examples

- Desktop connected via LAN cable
- Office wired networks

## Speed

- 10 Mbps (basic Ethernet)

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# Fast Ethernet

**Fast Ethernet** is a faster version of normal Ethernet used in local area networks (LAN).
Same network type, same cables — but higher speed

**How It Works**

- Same working as Ethernet
- Faster data transfer

**Speed**

- **100 times better then simple Ethernet**

**Examples**

Small office networks

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# Metro Ethernet



**Metro Ethernet** extends Ethernet technology to a **city level network**.

Used to connect **multiple LANs across a city**.

**Key Features**

Covers **metropolitan (city) area**

Uses **fiber optic cables**

Very high speed

Used by **large organizations**

**Speed**

**Example**

Connecting company branches in one city

University campuses across a city

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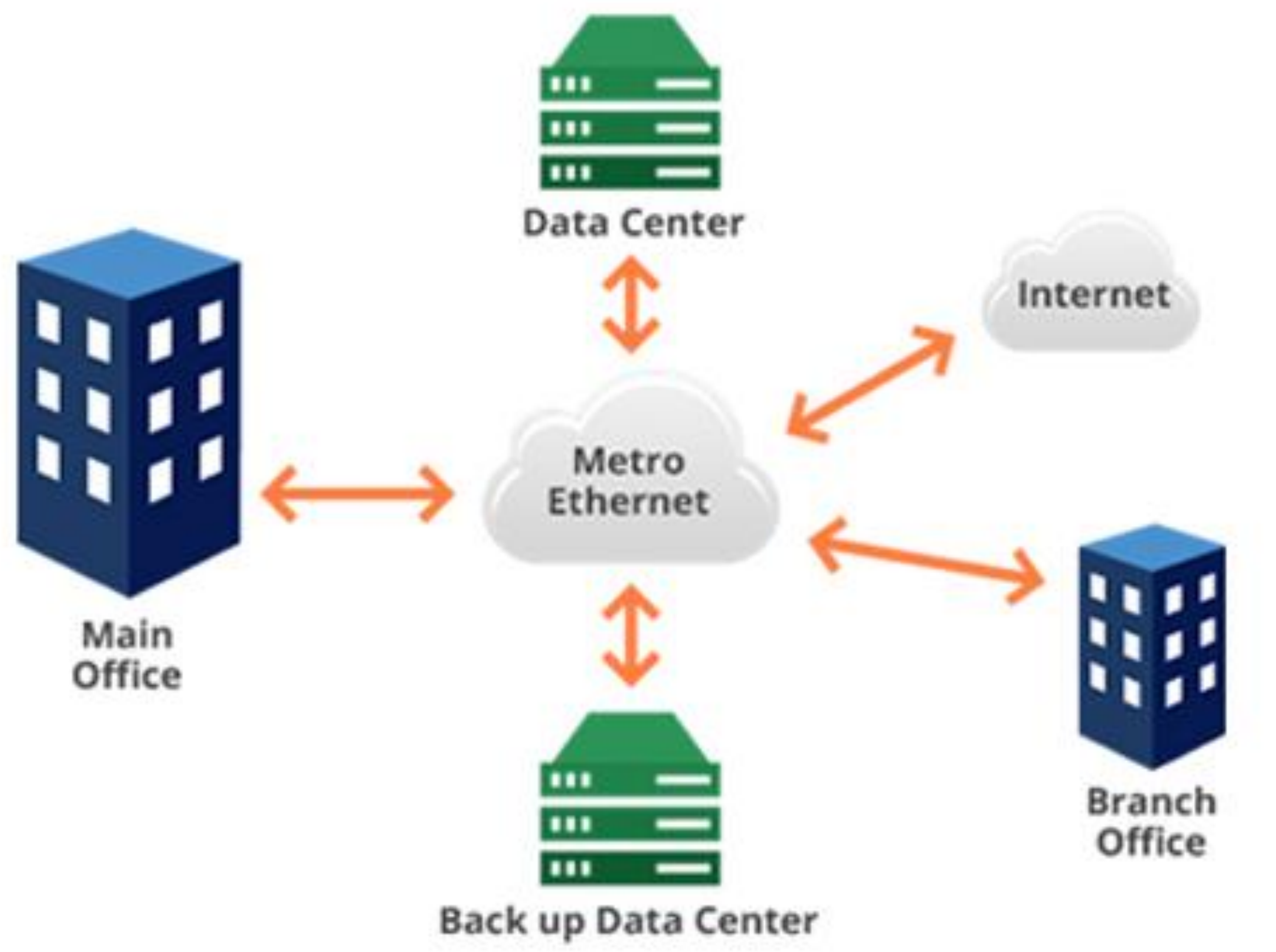
# Metro Ethernet



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# Comparasion Table

| Feature | Ethernet | Fast Ethernet | Metro Ethernet |
|---|---|---|---|
| Speed | 10 Mbps | 100 Mbps | 1–10 Gbps |
| Area | LAN (small) | LAN (small/medium) | City (MAN) |
| Cable | Copper | Copper | Fiber optic |
| Cost | Low | Medium | High |
| Usage | Old LANs | Modern LANs | City-wide networks |

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# Internetwork

An **internetwork** is formed by **connecting two or more networks**.

Internet (largest internetwork)

Company LAN + WAN + MAN combined

The **Internet** itself is the largest internetwork

(because it connects millions of networks worldwide)

**Example**

Your home Wi-Fi is one network

Google servers are another network

When you open Google:

Your home network connects to another network

This connection = **Internetwork**

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# THANK YOU

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